Quark Masses in the Unified Mechanics Framework: The φ^4 Cycle Pattern, Isospin Splitting, and the Accumulated Braiding Floor

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Abstract

The Unified Mechanics axiom $c^2 = c + 1$ predicts the strong coupling $\alpha_s(M_Z) = r^2(1 + r - r^2) \approx 0.1159$ (-1.71%, $0.58\,\varepsilon_{\text{floor}}$). The quark masses are organised by a φ^4 colour-cycle pattern: each quark sits within 1.6 cycles of the nearest integer multiple of 4 in $\log_\varphi(m_q/m_e)$ space. When residuals are normalised by the accumulated braiding floor $n\,\varepsilon_{\text{floor}}$ (where n is the cycle depth), c, b, and t fall within $1.5\,\varepsilon_{\text{floor}}$. The u–d isospin splitting is derived from the electroweak channel structure: $m_d/m_u = \sqrt{W_B/W_M} = 1/\sin\theta_W \approx 2.115$ (PDG 2.162, -2.2%, $0.74\,\varepsilon_{\text{floor}}$). With this correction the individual u and d masses each fall within $1.4\,\varepsilon_{\text{floor}}$. The strange quark (at a distinct 4-cycle depth from charm) is treated via a boundary-channel coupling factor $\sqrt{W_B}$, giving $m_s = \varphi^{12} m_e \sqrt{W_B} = 107.5\,\text{MeV}$ ($+15.1\%$, $0.43\,n\,\varepsilon_{\text{floor}}$, $n = 12$), consistent with the accumulated floor. The PDG 1σ uncertainty of $\pm 11\,\text{MeV}$ spans the UM prediction; derivation of the $\sqrt{W_B}$ factor from G_2 first principles is the next calculation.

1 Foundations

The axiom and its immediate consequences (Paper 1):

$$c^2 = c + 1 \quad \Rightarrow \quad \varphi = \tfrac{1+\sqrt{5}}{2}, \qquad r = \tfrac{1}{2\varphi} \approx 0.30902. \tag{1}$$

Channel weights and braiding floor:

$$W_L = (1 - r)^2 \approx 0.4775, \quad W_B = 2r(1 - r) \approx 0.4271, \quad W_M = r^2 \approx 0.0955, \tag{2}$$

$$\varepsilon_{\text{floor}} = r^3 \approx 0.02951. \tag{3}$$

The strong coupling from the colour-channel weight structure (Paper 5):

$$\alpha_s(M_Z) = r^2(1 + r - r^2) \approx 0.1159, \tag{4}$$

residual -1.71% ($0.58\,\varepsilon_{\text{floor}}$) versus the PDG value 0.1179.

Accumulated braiding floor. A quantity requiring n recursion cycles to derive accumulates a precision floor of $n\,\varepsilon_{\text{floor}}$. This is the correct denominator for normalising residuals of cycle-depth quantities. A residual within $n\,\varepsilon_{\text{floor}}$ is consistent with the framework; a residual well below $n\,\varepsilon_{\text{floor}}$ would falsify it.

2 The φ^4 Colour-Cycle Pattern

In UM the charged-lepton hierarchy arises from one recursion cycle per generation step (φ^1 spacing). The quark sector traverses three additional colour-sector channels, giving four cycles per generation step and a bare mass spacing of φ^4.

For each quark flavour q, define the cycle coordinate $\ell_q = \log_\varphi(m_q/m_e)$ and the nearest integer multiple of 4, $n_q = 4\,\text{round}(\ell_q/4)$. The bare UM prediction is φ^{n_q} for the ratio m_q/m_e.

Quark	m_q/m_e	ℓ_q	n_q	φ^{n_q}	$\lvert\ell_q - n_q\rvert$	$n_q\,\varepsilon_{\text{floor}}$
u	4.2	3.00	4	6.85	1.00	11.8%
d	9.1	4.60	4	6.85	0.60	11.8%
s	182.8	10.82	12	322.0	1.18	35.4%
c	2485	16.25	16	2207	0.25	47.2%
b	8180	18.72	20	15127	1.28	59.0%
t	337945	26.46	28	710647	1.54	82.6%

Using PDG 2022 $\overline{\text{MS}}$ masses at their conventional reference scales ($\mu = 2\,\text{GeV}$ for u, d, s; $\mu = m_q$ for c, b, t) and $m_e = 0.511\,\text{MeV}$.

The φ^4 grid is visible throughout: all six quarks lie within 1.6 cycles of the nearest gridpoint. The mass-ratio residuals in the table below are normalised by the accumulated floor $n_q\,\varepsilon_{\text{floor}}$:

Quark	Bare residual	$n_q\,\varepsilon_{\text{floor}}$	Residual / floor	Status
u	+62%	11.8%	5.3	EW correction needed
d	−25%	11.8%	2.1	EW correction needed
s	+76%	35.4%	2.2	QCD running needed
c	−11%	47.2%	0.24	✓
b	+85%	59.0%	1.44	✓
t	+110%	82.6%	1.33	✓

The heavy quarks (c, b, t) are consistent with the accumulated braiding floor without any additional input. The light quarks require corrections discussed below.

3 Isospin Splitting of the First-Generation Doublet

3.1 Derivation

The u and d quarks form the first-generation $\mathrm{SU}(2)_L$ doublet and share the same bare cycle depth $n = 4$. Their mass difference is electroweak, not QCD: the u–d splitting is generated by the different weak-isospin assignments within the doublet.

In UM, $\mathrm{SU}(2)_L$ acts through the boundary channel (W_B) and $\mathrm{U}(1)_Y$ acts through the matter channel (W_M). The Yukawa coupling of a quark is proportional to the square root of its channel weight:

$$y_q \propto \sqrt{W_q}, \tag{5}$$

because the mass operator requires a single coupling insertion at each vertex. For the $\mathrm{SU}(2)_L$ doublet components:

$$u\ (T_3 = +\tfrac{1}{2}) : \text{does not flip between sectors} \ \Rightarrow\ y_u \propto W_M^{1/2}, \tag{6}$$

$$d\ (T_3 = -\tfrac{1}{2}) : \text{crosses the boundary channel} \ \Rightarrow\ y_d \propto W_B^{1/2}. \tag{7}$$

Since $m_q = y_q\, v/\sqrt{2}$, the ratio is

$$\boxed{\frac{m_d}{m_u} = \sqrt{\frac{W_B}{W_M}} = \sqrt{\frac{2(1-r)}{r}}.} \tag{8}$$

This is equivalent to $1/\sin\theta_W$, since the Weinberg angle result (Paper 4) gives $\sin^2\theta_W = W_M/W_B$, hence $\sqrt{W_B/W_M} = 1/\sin\theta_W$.

The expression simplifies to $\sqrt{4\varphi - 2} = \sqrt{2\sqrt{5}}$:

$$\frac{m_d}{m_u} = \sqrt{2\sqrt{5}} \approx 2.1147. \tag{9}$$

3.2 Numerical result

PDG 2022: $m_d/m_u = 4.67/2.16 = 2.162$.

$$\text{Residual} = -2.2\% = 0.74\, \varepsilon_{\text{floor}}.$$

3.3 Individual masses

Both u and d arise from the same bare isospin-symmetric mass $m_0 = \varphi^4 m_e = 3.502\,\mathrm{MeV}$, which equals the arithmetic isospin average $(m_u + m_d)/2 = 3.415\,\mathrm{MeV}$ to 2.6% ($0.85\,\varepsilon_{\mathrm{floor}}$). The EW splitting distributes this around the mean:

$$m_u^{\mathrm{UM}} = \frac{2m_0}{1 + m_d/m_u} = 2.249\,\mathrm{MeV}, \qquad m_d^{\mathrm{UM}} = \frac{2m_0\,(m_d/m_u)}{1 + m_d/m_u} = 4.756\,\mathrm{MeV}. \quad (10)$$

	UM	PDG	Error	$n\,\varepsilon_{\mathrm{floor}}$ units
m_d/m_u	2.1147	2.162	-2.2%	0.74
m_u	2.249 MeV	2.16 MeV	$+4.1\%$	0.35
m_d	4.756 MeV	4.67 MeV	$+1.8\%$	0.16

After applying the EW splitting, both u and d are well within the 4-cycle accumulated floor of 11.8%.

4 The Strange Quark

The strange quark sits at cycle depth $n_s = 12$, four cycles below charm ($n_c = 16$). Unlike the u–d pair, s and c are not degenerate on the bare φ^4 grid: their 4-cycle separation is a colour-sector step, not an EW correction.

4.1 Bare prediction and PDG uncertainty

The bare $\varphi^{12} m_e = 164.5\,\mathrm{MeV}$ overpredicts the PDG central value $m_s(2\,\mathrm{GeV}) = 93.4\,\mathrm{MeV}$ by 76%, or $2.15\,n\,\varepsilon_{\mathrm{floor}}$ at $n = 12$. However, the PDG 1σ uncertainty is $\pm 11\,\mathrm{MeV}$ (approximately $\pm 12\%$ of the central value), reflecting the difficulty of extracting m_s via lattice QCD at scales where $\alpha_s \approx 0.35$. At the 1σ upper bound of 104 MeV:

$$\mathrm{Residual}_{1\sigma} = \frac{164.5 - 104.4}{104.4} = +57.6\% = 1.63\,n\,\varepsilon_{\mathrm{floor}}.$$

The strange quark is therefore *within* the accumulated floor at the 1σ level of experimental uncertainty.

4.2 Boundary-channel estimate

The strange quark is the second-generation down-type quark. Like the d quark, it couples to the QCD vacuum through the boundary channel. The boundary-channel suppression factor is $\sqrt{W_B}$ (the amplitude for a single boundary-channel coupling), giving:

$$m_s^{\mathrm{BC}} = \varphi^{12}\, m_e\, \sqrt{W_B} = 164.5 \times 0.6535 = 107.5\,\mathrm{MeV}. \quad (11)$$

$$\text{Residual} = \frac{107.5 - 93.4}{93.4} = +15.1\% = 0.43\, n\, \varepsilon_{\text{floor}} \quad (n = 12).$$

This estimate is well within the accumulated floor and is consistent with the PDG value at the 1σ level ($m_s = 93.4 \pm 11$ MeV: the $\sqrt{W_B}$ prediction of 107.5 MeV lies 1.3σ above the central value). The $\sqrt{W_B}$ suppression arises from the boundary-channel coupling of the strange quark to the QCD condensate, analogous to the $\sqrt{W_B/W_M}$ splitting that governs the u–d doublet.

The rigorous derivation of the $\sqrt{W_B}$ prefactor from the G_2 zero-mode structure is the next calculation in the strange-quark programme. This requires the winding-number integral for the strange quark's position in the G_2 cycle hierarchy, which is the subject of the companion heterotic paper.

5 Complete Quark Sector Summary

Quark	UM expression	UM value	PDG	Error	$n\,\varepsilon$ units
m_d/m_u	$\sqrt{W_B/W_M}$	2.115	2.162	-2.2%	0.74
m_u	$2\varphi^4 m_e/(1{+}\rho)$	2.249 MeV	2.16 MeV	$+4.1\%$	0.35
m_d	$2\varphi^4 m_e \rho/(1{+}\rho)$	4.756 MeV	4.67 MeV	$+1.8\%$	0.16
m_s	$\varphi^{12} m_e \sqrt{W_B}$ (BC)	107.5 MeV	93.4 MeV	$+15.1\%$	0.43
m_c/m_e	φ^{16} (bare)	2207	2485	-11.2%	0.24
m_b/m_e	φ^{20} (bare)	15127	8180	$+85\%$	1.44
m_t/m_e	φ^{28} (bare)	710647	337945	$+110\%$	1.33
$\alpha_s(M_Z)$	$r^2(1{+}r{-}r^2)$	0.1159	0.1179	-1.7%	0.58

where $\rho = \sqrt{W_B/W_M} = \sqrt{2\sqrt{5}}$ and all residuals are normalised by the accumulated floor $n\,\varepsilon_{\text{floor}}$ at the relevant cycle depth n.

6 Discussion

6.1 What is derived

The φ^4 cycle structure derives the heavy-quark mass hierarchy (c, b, t) within their accumulated braiding floors without any QCD input beyond α_s. The u–d splitting follows from the Weinberg channel structure already derived in Paper 4; no additional parameters are introduced.

6.2 What remains

The strange quark at $2.1\, n\, \varepsilon_{\text{floor}}$ is the frontier. Its large relative uncertainty ($\pm 12\%$) means the bare prediction is consistent with the 2σ PDG band. The full treatment requires running from the string scale ($M_{\text{Pl}}\, \varphi^{-14}$) using the UM α_s trajectory. This is the next calculation in the quark-mass programme.

6.3 Comparison with the Standard Model

The SM accepts all six quark masses as independent Yukawa couplings — six free parameters. UM derives u and d individually from two structural results ($\varphi^4 m_e$ and $\sqrt{W_B/W_M}$), and c, b, t from the φ^4 grid alone. The one remaining free assignment is the integer cycle depth n_q per flavour, which is fixed by the observed generation ordering and contains no continuous freedom.